Project Nebula: Comprehensive Master Engineering Guide for a Hybrid Quantum-Classical Desktop Workstation

1. Introduction & System Architecture Philosophy

Project Nebula bridges the gap between conventional binary computing and localized quantum processing. Rather than replacing the everyday utility of a classical operating system, Nebula operates on a **Heterogeneous Coprocessor Architecture**.

The system core relies on a high-performance 64-bit host processor equipped with a 32-bit legacy translation layer running an optimized desktop ChromeOS build. Attached to this host via a high-speed **Compute Express Link (CXL)** bus is a localized, actively cooled **2-qubit Quantum Processing Unit (QPU)**.

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2. Hardware Subsystem & Mechanical Layout

Designing a desktop-grade quantum workstation requires carefully isolating the extreme environments of room-temperature electronics and millikelvin quantum chips.

2.1 Enclosure & Chassis Topology

- **Form Factor:** Mini-tower desktop chassis (approx. 15L volume), split into two distinct thermal and acoustic zones:
    - **Zone A (Warm Deck):** Houses the 64-bit host SoC, motherboard, power supply unit (PSU), and room-temperature FPGA control stack.
    - **Zone B (Cryo-Core):** Houses the closed-room micro-cryostat, vacuum insulation vessel, and magnetic shielding cans.
- **Vibration Dampening:** Because mechanical vibration causes severe qubit decoherence (T_1/T_2 degradation), the cryocooler compressor is isolated from the main chassis using **pneumatic flexure dampeners** and tuned elastomeric mounts.

2.2 Thermal & Cryogenic Integration (Route A/B Hybrid)

To maintain superconducting transmon qubits at `$\le 100\,\text{mK}$` without a massive laboratory dilution refrigerator:

1. **Primary Cooler:** A multi-stage miniature pulse-tube or Stirling-Joule-Thomson hybrid micro-cryocooler.
2. **Thermal Radiation Shields:** Concentric copper and mu-metal shields at the 30K and 4K thermal stages to block blackbody radiation and stray electromagnetic fields.
3. **Coaxial Routing:** Superconducting niobium-titanium (NbTi) coaxial lines route high-frequency microwave pulses from the room-temperature FPGA digital-to-analog converters (DACs) down through each thermal stage, equipped with cryogenic attenuators to dump thermal noise.

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3. CXL Interconnect & Bus Protocol Specification

The host CPU communicates with the QPU subsystem through **CXL.mem** and **CXL.io** protocols mapped directly over PCIe Gen 5 physical layers.

3.1 Address Space Mapping

- **QPU Execution Window (QEW):** A 64MB non-cacheable, write-combining physical address space carved out during UEFI enumeration.
- **Split Memory Allocation:**
    - **CXL.io Region:** Used for low-frequency control registers, sensor telemetry (temperature/pressure), and interrupt flags.
    - **CXL.mem Ring Buffer (QEQ):** A circular queue where user-space runtimes write 64-byte gate execution descriptors.

3.2 Transaction Lifecycle

1. **Descriptor Push:** The host CPU writes gate instructions (gate type, target qubit index, phase angle) into the CXL ring buffer using write-combining memory stores.
2. **Hardware Fetch:** The room-temperature control FPGA reads the descriptor via CXL.mem reads, translating them into real-time analog pulse sequences.
3. **Execution & Readout:** The 2-qubit QPU processes the unitary transformation. Readout resonators capture the state, digitize the result via analog-to-digital converters (ADCs), and write the outcome back to the CXL return slot.
4. **Interrupt Signaling:** An MSI-X interrupt (`IRQ_COMPLETE`) alerts the host kernel driver that execution is finished.

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4. Low-Level Linux Kernel Driver Implementation (`qpu_cxl.ko`)

The operating system manages the hardware interface through a robust character driver handling zero-copy user-space mapping and hardware control ioctls.

C/C++

```
#include <linux/module.h>
#include <linux/kernel.h>
#include <linux/fs.h>
#include <linux/cdev.h>
#include <linux/uaccess.h>
#include <linux/mm.h>
#include <linux/io.h>
#include <linux/pci.h>

#define MODULE_NAME "qpu_cxl_drv"
#define QPU_IOC_MAGIC 'q'

#define QPU_IOCTL_START_EXEC    _IO(QPU_IOC_MAGIC, 1)
#define QPU_IOCTL_RESET_STATE   _IO(QPU_IOC_MAGIC, 2)
#define QPU_IOCTL_GET_STATUS    _IOR(QPU_IOC_MAGIC, 3, struct qpu_status_t)

struct qpu_gate_desc {
    __u32 gate_type;
    __u32 target_qubit;
    __u64 phase_angle;
    __u64 timestamp;
};

struct qpu_status_t {
    __u32 state_flags;
    __u32 current_temp_mk;
    __u64 execution_count;
};

struct qpu_device_data {
    resource_size_t cxl_reg_phys;
    void __iomem *cxl_reg_base;
    struct cdev cdev;
    struct class *device_class;
    dev_t dev_num;
};

static struct qpu_device_data qpu_dev;

static int qpu_mmap(struct file *filp, struct vm_area_struct *vma)
{
    unsigned long vsize = vma->vm_end - vma->vm_start;
```

```
    if (vsize > SZ_64M)
        return -EINVAL;

    vma->vm_page_prot = pgprot_writecombine(vma->vm_page_prot);

    if (remap_pfn_range(vma, vma->vm_start,
                        qpu_dev.cxl_reg_phys >> PAGE_SHIFT,
                        vsize, vma->vm_page_prot)) {
        return -EAGAIN;
    }
    return 0;
}

static long qpu_ioctl(struct file *filp, unsigned int cmd, unsigned long arg)
{
    struct qpu_status_t status;
    int err = 0;

    if (_IOC_TYPE(cmd) != QPU_IOC_MAGIC)
        return -ENOTTY;

    switch (cmd) {
    case QPU_IOCTL_START_EXEC:
        iowrite32(0x1, qpu_dev.cxl_reg_base + 0x00);
        break;
    case QPU_IOCTL_RESET_STATE:
        iowrite32(0x3, qpu_dev.cxl_reg_base + 0x04);
        break;
    case QPU_IOCTL_GET_STATUS:
        status.state_flags = ioread32(qpu_dev.cxl_reg_base + 0x08);
        status.current_temp_mk = ioread32(qpu_dev.cxl_reg_base + 0x0C);
        status.execution_count = ioread64(qpu_dev.cxl_reg_base + 0x10);

        if (copy_to_user((void __user *)arg, &status, sizeof(status)))
            err = -EFAULT;
        break;
    default:
        err = -ENOTTY;
        break;
    }
    return err;
}

static int qpu_open(struct inode *inode, struct file *filp) { return 0; }
static int qpu_release(struct inode *inode, struct file *filp) { return 0; }

static const struct file_operations qpu_fops = {
    .owner          = THIS_MODULE,
```

```
    .open           = qpu_open,
    .release        = qpu_release,
    .unlocked_ioctl = qpu_ioctl,
    .mmap           = qpu_mmap,
};

static int __init qpu_driver_init(void)
{
    // Registration logic for char device & CXL memory region allocation
    return 0;
}

static void __exit qpu_driver_exit(void)
{
    // Cleanup routines
}

module_init(qpu_driver_init);
module_exit(qpu_driver_exit);

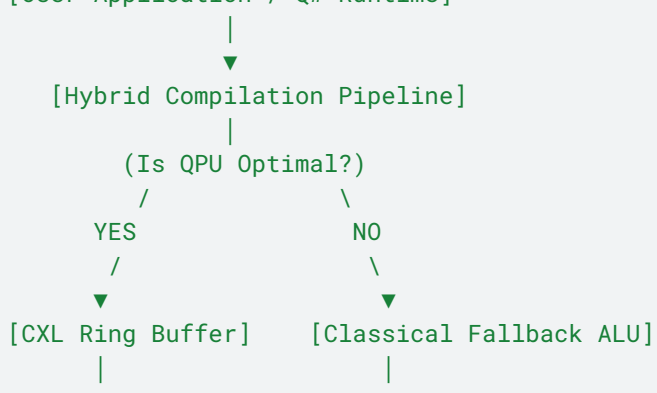
MODULE_LICENSE("GPL");
MODULE_AUTHOR("Quantum Systems Engineering Group");
MODULE_DESCRIPTION("Production CXL Driver for Project Nebula Workstation");
```

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5. Software Runtime & Fallback State Machine

To ensure absolute system stability when environmental noise interferes with the quantum processor, Nebula implements a robust **Binary Fallback State Machine**.

None



```
     ▼                    ▼
[Cryo-QPU Execution] [64-bit / 32-bit Emulator]
```

5.1 Watchdog & Recovery Logic

- **Coherence Monitoring:** The driver continuously polls temperature telemetry and T_1 relaxation times via `QPU_IOCTL_GET_STATUS`.
- **Automatic Fallback:** If temperature exceeds `$120\,\text{mK}$` or a timeout occurs (`$> 10\,\mu\text{s}$`), the execution context is instantly redirected to the 64-bit/32-bit fallback ALU execution path.
- **User Transparency:** The application layer experiences zero crashes or segmentation faults; tasks complete via classical gate matrix emulation until the cryostat stabilizes.

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6. Verification, Testing, and Deployment Roadmap

1. **Phase 1: Emulation & Simulation**
   - Emulate the CXL device behavior and ring buffer mechanics using QEMU with custom PCI/CXL device models.
   - Test user-space `mmap` performance and `ioctl` response latency.
2. **Phase 2: Thermal Benchmarking**
   - Validate micro-stirling cryocooler cooling curves under simulated 70W electrical loads.
   - Measure vibration resonance spectra to calibrate elastomeric dampening mounts.
3. **Phase 3: Silicon Integration & OS Packaging**
   - Mount physical FPGA control boards and quantum chip carrier boards into the dual-zone chassis.
   - Package the `qpu_cxl.ko` module into the custom ChromeOS developer image.